CallAppoint

Clinical Research System Handbook

A Comprehensive Guide to the Research-Oriented Clinical Database

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1. The Problem: Unstructured Clinical Data

The "Monkey" Explanation

Imagine a library where every book is just a pile of random loose papers thrown into a box. If you want to find out how many books mention "Apples," you have to read every single page in every single box. That's how traditional medical apps work—they just store big text notes or PDF files.

What we faced

In the original **CallAppoint** system:

• **Medical Reports** were just PDF links.
• **Diagnoses** were just a single text box where doctors could write anything they wanted.
• **Biomarkers** (like blood sugar or tumor levels) were hidden inside PDFs.

The Research Gap

Researchers couldn't answer simple questions like:

• *"How many Stage III cancer patients are smokers?"*
• *"Does this specific genetic mutation respond better to the treatment?"*

Computers can't "read" PDFs and random notes easily. We needed **Structured Data**.

2. Our Approach vs. Actual Solution

The Vision

We wanted a system that was easy for doctors to use but "Gold Mine" for researchers.

How we approached it

Initially, we thought about just adding more text fields. But we realized that for true research (especially in cancer), we needed a **Categorical System**.

How we actually solved it

We built a **Layered Architecture** across 5 checkpoints:

1. **Background Layer:** Lifestyle & History.
2. **Clinical Layer:** ICD-10 Coding & TNM Staging.
3. **Temporal Layer:** Time-series Biomarkers.
4. **Precision Layer:** Genomic/Genetic Variants.
5. **Insight Layer:** Analytics Dashboards.

We enforced **Security at the API level** so that doctors only see their own patients, and patients can't fake their own genetic results.

3. The 5 Checkpoints & Their Necessity

Checkpoint 1: The Patient Baseline (Lifestyle)

• **What:** Smoking, Alcohol, Family History.
• **Why:** Researchers need to identify "Risk Factors." You can't study lung cancer without knowing if the patient smokes.

Checkpoint 2: The Diagnosis Engine (ICD & TNM)

• **What:** Universal codes (ICD-10) and Cancer Staging (TNM).
• **Why:** Every hospital in the world uses these. It makes our data "Interoperable."

Checkpoint 3: The Biomarker Tracker

• **What:** Numeric values (e.g., PSA: 4.5 ng/mL).
• **Why:** Charts! We can now draw a line graph showing a patient getting better over time.

Checkpoint 4: The Precision Layer (Genomics)

• **What:** Gene mutations (e.g., BRCA1).
• **Why:** This is the future of medicine. We can study "Targeted Therapies."

Checkpoint 5: Analytics Dashboards

• **What:** Aggregated charts for the whole hospital.
• **Why:** It gives the hospital "Big Picture" insights immediately.

4. User Guide: How to use the System

For Doctors (The Scientist)

Doctors no longer just "write notes." They "record data."

1. **During Appointment:** The doctor opens the patient's Clinical Profile.

2. **Diagnosis:** They pick an ICD-10 code and select the T, N, and M stages from dropdowns.

3. **Lab Results:** Instead of just looking at the PDF, they type in the numeric value (e.g., "Creatinine: 1.2").

4. **Genetics:** They record any mutations found in the lab.

For Patients (The Participant)

1. **My Journey:** Patients see a "Health Trend" graph instead of just a list of files.

2. **Profile:** They fill in their lifestyle habits once, so every doctor they see already knows their history.

3. **Safety:** They can see their genetic results, but they cannot edit them—ensuring the record remains official.

For Researchers (The Analyst)

1. **Analytics:** They hit the `/analytics/summary/` endpoint.

2. **Output:** They get a breakdown of all stages and genetic variants across the entire patient population instantly.

5. Future Enhancements & Current Loopholes

Current Loopholes (Things to watch out for)

1. **Manual Entry Fatigue:** Doctors hate typing. If we don't build a very fast UI with autocompletes, they might just go back to writing text notes.

2. **Data Silos:** This data is only inside CallAppoint. We need to implement FHIR (Fast Healthcare Interoperability Resources) to export it to other hospitals.

3. **Verification:** Currently, any doctor can enter genetic data. In a real hospital, only a Pathologist or Geneticist should be able to do this.

Future Enhancements

1. **AI Diagnosis Support:** Use the structured data to suggest a diagnosis to the doctor.

2. **Automated PDF Parsing:** Use OCR/AI to read the lab PDFs and automatically fill in the Biomarker table so the doctor doesn't have to type.

3. **Research Export:** A button to export the entire database to a CSV/Excel file for use in tools like R or Python for medical statistics.